SI 201 Discussion 10

JSON

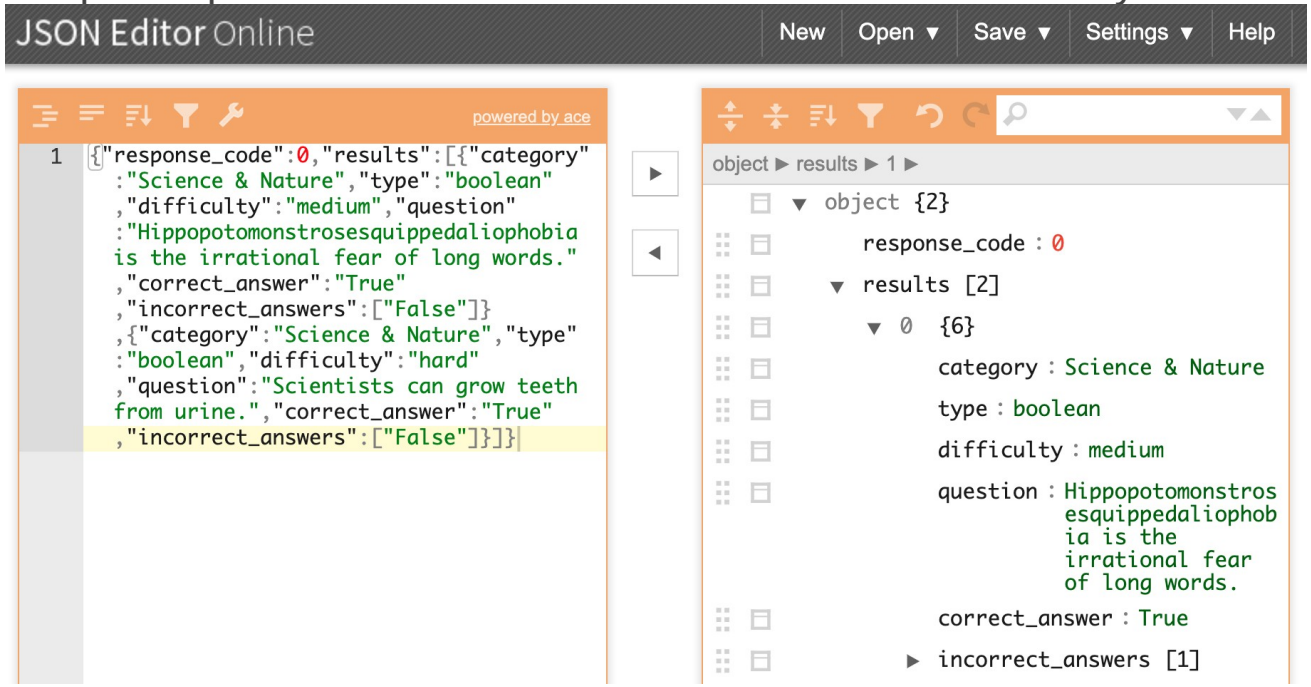
JSON Workflow

- Import the JSON module
  - `import json`
- Get the file object – open the file for reading
  - `source_dir = os.path.dirname(__file__)`
  - `full_path = os.path.join(source_dir, 'com.json')`
  - `file = open(full_path, 'r')`
- Read the contents into one big string
  - `contents = file.read()`
- Close the file
  - `file.close()`
- Convert the string into a Python object (list or dictionary)
  - `data = json.loads(string)`
- Process the Python object - data
- Remember to use null instead of None

JSON Editor

JSON Editor Online : https://jsoneditoronline.org

It’s super helpful to see the structure in a readable format that you can manipulate



Today’s Task pt.1

Use the provided JSON document (books.json)

TASK 1: Write a function `read_json(file)` that reads a JSON document, decodes the file content, and returns a dictionary.

TASK 2: Write a function `get_shortest_book(data)` that returns the title of the book with the fewest pages.
`e.g. ‘Infinite Jest’`

Today’s Task pt.2

Use the provided JSON document (books.json)

TASK 3: Write a function `title_by_year(year, data)` that returns a dictionary where the keys are authors and the values are their books, each book should be created in the year.

```
e.g. {'Thomas Pynchon': ["Gravity's Rainbow (Penguin Classics Deluxe Edition)"], 'Gabriel García Márquez': ['One hundred years of solitude'], 'Kazuo Ishiguro': ['Never Let Me Go']}
```

TASK 4: Write a function `publisher_by_letter(letter, data)` that returns a dictionary where the keys are publisher whose name begins with the specified letter and the values are their the title of the books

```
e.g. {'Penguin Classics': ["Gravity's Rainbow (Penguin Classics Deluxe Edition)"], 'Penguin Publishing Group': ['Bunny'], 'Penguin Books': ['Eileen', 'White noise'], 'Penguin Press': ['Eileen']}
```